MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting

Linux Lab: D1_AWK

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Part A - Numerical and String Comparisons

1. Display the name, department, and salary of employees earning more than 70000.

awk '$4 > 70000 {print $2, $3, $4}' employee.txt

2. Display employees whose salary is between 60000 and 80000, including both limits.

awk '$4 >= 60000 && $4 <= 80000' employee.txt

3. Display employees whose employment status is Inactive.

awk '$5 == "Inactive"' employee.txt

4. Display employees belonging to the VLSI department.

awk '$3 == "VLSI"' employee.txt

Part B - Regular-Expression Patterns

1. Display all records containing the word Security.

awk '/Security/' employee.txt

2. Display employees whose names begin with the letters A, B, or C.

Apply the regular expression only to the employee-name field.

awk '$2 ~ /^[ABC]/' employee.txt

3. Display employees whose department field does not match Security.

awk '$3 !~ /^Security$/' employee.txt

Part C - Logical Patterns

1. Display employees who work in the Security department and earn more than 70000.

awk '$3 == "Security" && $4 > 70000' employee.txt

2. Display employees who belong to either the VLSI or Networking department.

awk '$3 == "VLSI" || $3 == "Networking"' employee.txt

3. Display employees who: Work in the VLSI department, earn more than 70000, have Active status

awk '$3 == "VLSI" && $4 > 70000 && $5 == "Active"' employee.txt

Part D - Conditional Pattern

1. Department-dependent salary condition

Apply different salary conditions:

If the employee belongs to Security, select employees earning more than 75000.

Otherwise, select employees earning more than 70000.

Display the employee name, department, and salary.

awk '($3 == "Security" ? $4 > 75000 : $4 > 70000) {print $2, $3, $4}' employee.txt

Part E - Grouped Patterns

1. Display active employees who belong to either the Security or VLSI department.

The Active condition must apply to both departments.

awk '$5 == "Active" && ($3 == "Security" || $3 == "VLSI")' employee.txt

2. Display employees who satisfy one of these conditions:

Work in Security and earn more than 80000

Work in VLSI and earn more than 75000

awk '($3 == "Security" && $4 > 80000) || ($3 == "VLSI" && $4 > 75000)' employee.txt

3. Display active employees who satisfy one of these conditions:

Work in Security and earn more than 70000

Work in VLSI and earn more than 75000

awk '$5 == "Active" && (($3 == "Security" && $4 > 70000) || ($3 == "VLSI" && $4 > 75000))' employee.txt

Part F - Negated Patterns

1. Display employees whose status is not Active.

Use an exact field comparison because the word Inactive contains the text Active.

awk '$5 != "Active"' employee.txt

2. Display employees who do not belong to either the Security or VLSI department.

awk '!($3 == "Security" || $3 == "VLSI")' employee.txt

Part G - Range Patterns

1. Display all records beginning with employee ID 103 and ending with employee ID 108. Include both boundary records.

awk '$1 == 103, $1 == 108' employee.txt

2. Display all records beginning with David and ending with Irene. Include both boundary records.

awk '$2 == "David", $2 == "Irene"' employee.txt

Part H - Built-In Variables

1. Display only the first five records. Include the record number in the output.

awk 'NR <= 5 {print NR, $0}' employee.txt

2. Display records numbered 4 through 9 using the built-in variable NR.

awk 'NR >= 4 && NR <= 9' employee.txt

3. Display only records containing exactly five fields. Print the record number and the complete record.

awk 'NF == 5 {print NR, $0}' employee.txt

4. Display only odd-numbered records using NR.

awk 'NR % 2 == 1' employee.txt

Part I - Arithmetic Conditions

1. Assume each employee receives a bonus equal to 10% of the salary. Display employees whose calculated bonus exceeds 7000.

Print:

Employee name

Salary

Calculated bonus

awk '$4 * 0.10 > 7000 {printf "%s %d %.2f\n", $2, $4, $4 * 0.10}' employee.txt

2. Question 24 Revised salary

Calculate the revised salary after adding a 12% increment. Display employees whose revised salary exceeds 85000.

awk '$4 * 1.12 > 85000 {printf "%s %d %.2f\n", $2, $4, $4 * 1.12}' employee.txt

3. Calculate the monthly salary by dividing the annual salary by 12. Display employees whose monthly salary exceeds 6000.

awk '$4 / 12 > 6000 {printf "%s %d %.2f\n", $2, $4, $4 / 12}' employee.txt

Part J - Integrated Challenge

1. Generate a report containing active employees who satisfy one of these conditions:

Work in Security and earn at least 70000

Work in VLSI and earn at least 75000

Work in Networking and earn at least 70000

The output must contain:

Employee ID

Employee name

Department

Salary

Status

Add a report heading using BEGIN and print the total number of selected employees using END.

```
awk 'BEGIN {
   print "ACTIVE EMPLOYEE REPORT"
   print "ID Name Department Salary Status"
}
$5 == "Active" && (
   ($3 == "Security" && $4 >= 70000) ||
   ($3 == "VLSI" && $4 >= 75000) ||
   ($3 == "Networking" && $4 >= 70000)
) {
   print $1, $2, $3, $4, $5
   count++
}
```

```
END {
    print "Total selected employees:", count
}' employee.txt
```